

CLONE-POS

Staff App — System, Technology & Feature Overview

Mobile-only · Offline-first · LAN-native

01 SUMMARY

Clone-POS is a mobile-only inventory-management and point-of-sale coordination system for multi-store retail and manufacturer field-sales. It is built for deterministic, offline-resilient operation: the store never stops functioning because the Master is unreachable. There is no web app and no AI in the product surface.

Devices take one of two roles from a single Staff App build. The Master (one per store, landscape tablet, fixed-geometry control panel) is the authoritative source of truth. Satellites (staff terminals) sell, look up and scan — online in real time, or queued locally when offline and reconciled on reconnect. A separate phone-only Clone-POS client is the customer-facing shopper.

Inventory truth is never overwritten. Every device appends timestamped events to a ledger; the Master replays them in order to compute real stock. Coordination between Master and Satellites runs entirely over the local network — presence and calling, hands-free voice, and catalog mirroring — with no cloud dependency for daily operation.

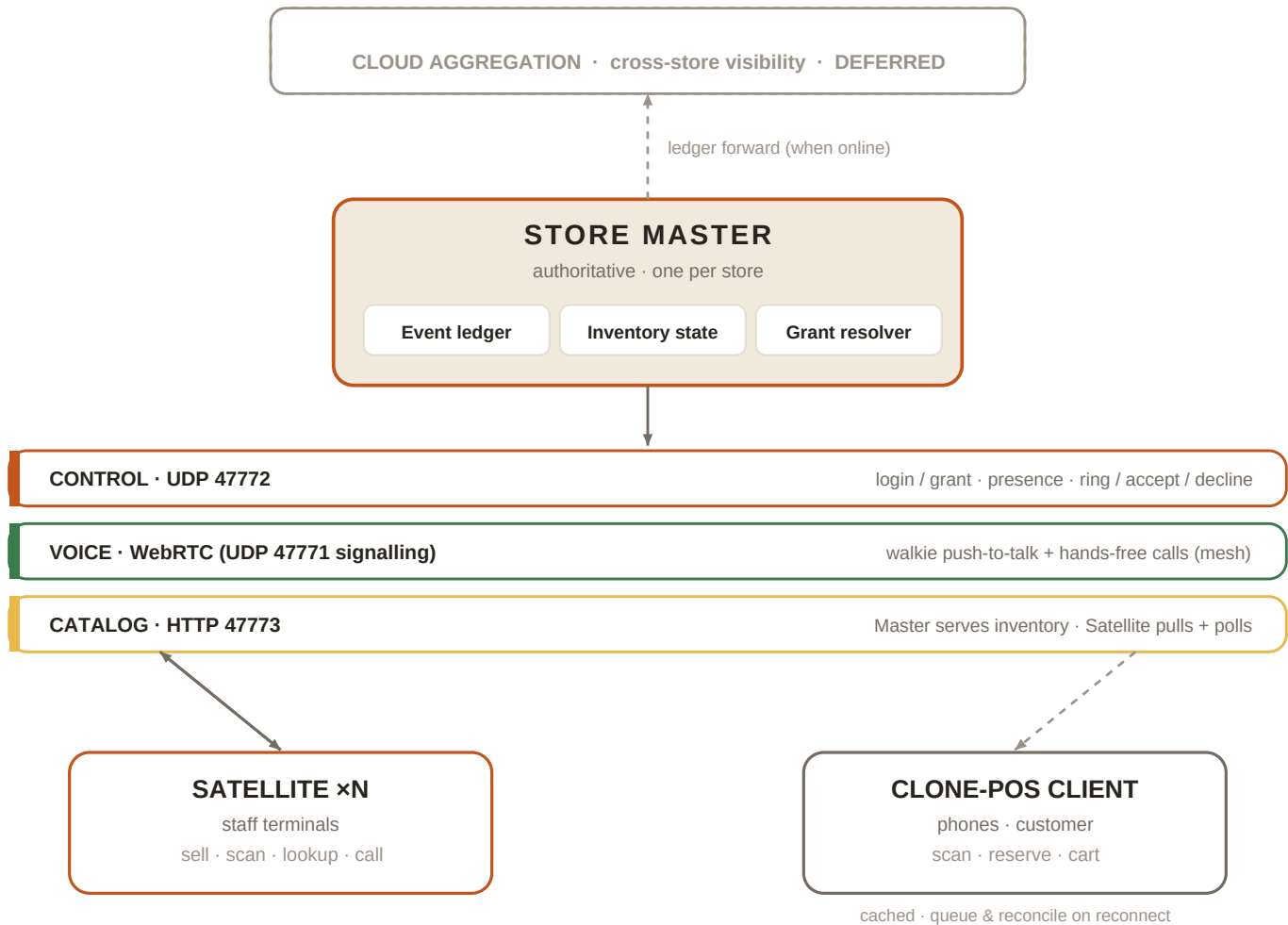
PLATFORM	SOURCE OF TRUTH	COORDINATION	RESILIENCE
Flutter / Dart	Event ledger	LAN mesh	Offline-first
Monorepo, Android tablets	HLC-timestamped, replayed	UDP + WebRTC + HTTP	Queue & reconcile

02 TECHNOLOGY STACK

Framework	Flutter (Dart) — single codebase, Master + Satellite roles chosen at pairing
Structure	Monorepo: packages/core (models, sync, repositories), packages/widgets, apps/staff_app
Voice / calling	flutter_webrtc — audio-only peer mesh; permission_handler; wakelock_plus
Signalling	dart:io RawDatagramSocket — UDP broadcast on the LAN (no server)
Audio cues	audioplayers — ring / accept / decline tones
Data import	csv · excel · xml · archive · file_picker — bulk catalog import
Media / files	image_picker · path_provider — product photos, local persistence
Iconography	material_symbols_icons; app font Google Sans Flex (fallback Roboto Flex)

03 SYSTEM ARCHITECTURE

One Master per store is authoritative. Satellites and clients operate on cached data and reconcile to it. Three independent LAN planes connect Master and Satellites; the cloud tier is for cross-store visibility only and is deferred.



READING THE DIAGRAM

Solid arrows are live LAN links between Master and Satellites; dashed arrows are deferred or cached-and-reconciled paths. The three planes are independent sockets running at once.

04 FLEET & ACCESS CONTROL (CLONES)

Per-clone capability

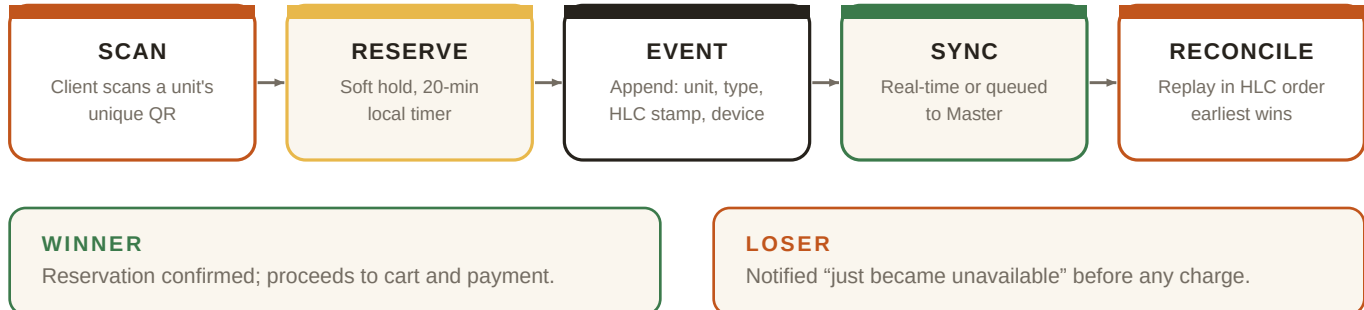
- Master grants each Satellite a named identity + department
- Feature-level access list resolved per clone
- Create / delete clones; live LINKED presence
- Grants pushed instantly when access changes

SaaS plan tiers

- Basic — up to 6 clones
- Standard — up to 12 clones
- Enterprise — up to 24 clones
- Capacity locking beyond the active plan

05 DATA FLOW — RESERVATION & LEDGER

The overbooking-prevention path. A scan creates a soft reservation against one physical unit — not a SKU count — timed on the customer's own device so it survives a Master that is offline for the whole window. The Master reconciles by replaying events in Hybrid Logical Clock order; the earliest reservation wins, and any loser is told before payment.



06 WHY HYBRID LOGICAL CLOCKS

Sorting reservations by raw wall-clock time let a device with a slow or wrong clock silently win every race. Each device now carries a Hybrid Logical Clock: a physical millisecond component plus a logical counter. On receiving any remote event, a device merges the remote clock, so its next timestamp is guaranteed later than anything it has already seen — the race is fair regardless of clock skew. Ties break by physical, then logical, then device id.

INVARIANT No device ever overwrites a stock number. State = ordered replay of the event ledger.

07 COMMUNICATION LAYER (LAN)

- **Presence & pairing** Satellite broadcasts login; Master answers with a signed grant and tracks live clones.
- **Calling** Ring / accept / decline both directions, 30-second ringtone, in-tree call banners.
- **Hands-free voice** WebRTC audio mesh; a connected call rides through weak-signal dropouts and self-recovers.
- **Catalog mirror** Master serves its inventory over HTTP; Satellites pull on link and poll a cheap version token.

08 FEATURE MAP

Master control panel

- Fixed 1280×800 four-region kiosk layout
- Stacked-card home: Active Carts, Clones, Inventory, Logistics, Analytics, Accounts, Sales Kit, Data
- Secondary index drives the Master display

Clones (fleet)

- Per-clone identity, department & feature grants
- Create / delete; SaaS tiers with capacity lock
- Live LINKED presence; instant grant push
- Hub & scrollable fleet grid

Selling & sync

- Offline sale, queued and reconciled on reconnect
- Event-sourced ledger, HLC ordering
- Soft reservation, earliest-timestamp wins
- Active carts, store-wide live view

Inventory

- Card / list / opened browser views
- Bulk import: CSV / Excel (IKEA, Shopify, Myntra)
- Product photos, categories, per-unit status
- Master-served catalog mirrored to Satellites

Communication

- LAN walkie-talkie, push-to-talk mesh
- Master ↔ Satellite calling, both directions
- Hands-free call voice with dropout recovery
- On-clone live link readout for range testing

Platform

- Light / dark app theme across every surface
- Role chooser at pairing (Master / Satellite)
- Business + clone identity persistence
- Wakelock during calls (MIUI Wi-Fi power-save)

09 DEFERRED — NOT BUILT AGAINST

🔗 HQ / cross-store cloud	Only internet-facing tier; under security review, may be dropped.
🔗 Label-printer brand	Zebra / Brother / TSC / Honeywell / Bixolon shortlist; final pick pending.
🔗 Data vault access gate	PIN / biometric required, sequenced to the final build stage.
🔗 Android TV / POS panels	Real future targets; after tablet + phone builds land.

DESIGN LANGUAGE

Mid-1970s plastic consumer electronics — Braun (Dieter Rams) and Bang & Olufsen: warm, tactile, restrained. Chunky rounded keys, recessed display windows, mustard + rust accents.